Year 11 Foundation

REVISION PACK

Handling Data 4

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

Q	Working	Answer
frequency tables		
1.1	$(a)\ 0:3,\ 1:8,\ 2:6,\ 3:2,\ 4:1$ $(c)\ 4-0=4$ $(d)\ 0\times3+1\times8+2\times6+3\times2+4\times1=30$ $30\div20=1.5$	**(a) frequencies 3, 8, 6, 2, 1** **(b) 1** **(c) 4** **(d) 1.5**
1.2	$(a)\ 0:2,\ 1:4,\ 2:8,\ 3:4,\ 4:2$ $(c)\ 4-0=4$ $(d)\ 0\times2+1\times4+2\times8+3\times4+4\times2=40$ $40\div20=2$	**(a) frequencies 2, 4, 8, 4, 2** **(b) 2** **(c) 4** **(d) 2**
1.3	$(a)\ 0:4,\ 1:9,\ 2:4,\ 3:2,\ 4:1$ $(c)\ 4-0=4$ $(d)\ 0\times4+1\times9+2\times4+3\times2+4\times1=27$ $27\div20=1.35$	**(a) frequencies 4, 9, 4, 2, 1** **(b) 1** **(c) 4** **(d) 1.35**
1.4	$(a)\ 0:2,\ 1:5,\ 2:7,\ 3:4,\ 4:2$ $(c)\ 4-0=4$ $(d)\ 0\times2+1\times5+2\times7+3\times4+4\times2=39$ $39\div20=1.95$	**(a) frequencies 2, 5, 7, 4, 2** **(b) 2** **(c) 4** **(d) 1.95**
1.5	$(a)\ 0:3,\ 1:9,\ 2:5,\ 3:2,\ 4:1$ $(c)\ 4-0=4$ $(d)\ 0\times3+1\times9+2\times5+3\times2+4\times1=29$ $29\div20=1.45$	**(a) frequencies 3, 9, 5, 2, 1** **(b) 1** **(c) 4** **(d) 1.45**
1.6	$(a)\ 0:5,\ 1:9,\ 2:4,\ 3:1,\ 4:1$ $(c)\ 4-0=4$ $(d)\ 0\times5+1\times9+2\times4+3\times1+4\times1=24$ $24\div20=1.2$	**(a) frequencies 5, 9, 4, 1, 1** **(b) 1** **(c) 4** **(d) 1.2**
pie charts from a table		

1.7	$360 \div 30 = 12$ $7 \times 12 = 84$	**84°**
1.8	$360 \div 24 = 15$ $10 \times 15 = 150$	**150°**
1.9	$360 \div 40 = 9$ $14 \times 9 = 126$	**126°**
1.10	$360 \div 18 = 20$ $5 \times 20 = 100$	**100°**
1.11	$360 \div 45 = 8$ $12 \times 8 = 96$	**96°**
1.12	$360 \div 60 = 6$ $22 \times 6 = 132$	**132°**
the median of six numbers		
1.13	$2, 4, 5, 7, 8, 10$ $(5 + 7) \div 2 = 6$	**6**
1.14	$3, 4, 6, 9, 12, 15$ $(6 + 9) \div 2 = 7.5$	**7.5**
1.15	$1, 3, 5, 6, 7, 9$ $(5 + 6) \div 2 = 5.5$	**5.5**
1.16	$6, 8, 10, 11, 12, 14$ $(10 + 11) \div 2 = 10.5$	**10.5**
1.17	$1, 2, 4, 5, 9, 9$ $(4 + 5) \div 2 = 4.5$	**4.5**
1.18	$11, 13, 15, 16, 18, 20$ $(15 + 16) \div 2 = 15.5$	**15.5**
misleading charts		
1.19		**The vertical axis does not start at 0 (it starts at 50), so the differences look bigger than they are.**

1.20		**The vertical axis does not start at 0 (it starts at 80), so the differences look bigger than they are.**
1.21		**The vertical axis does not start at 0 (it starts at 120), so the differences look bigger than they are.**
1.22		**The vertical axis does not start at 0 (it starts at 40), so the differences look bigger than they are.**
1.23		**The vertical axis does not start at 0 (it starts at 60), so the differences look bigger than they are.**
1.24		**The vertical axis does not start at 0 (it starts at 30), so the differences look bigger than they are.**
questionnaires		
1.25		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6; no box for 0**
1.26		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 7; no box for 0**
1.27		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6 KD; no box for 0**
1.28		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6; no box for 0**
1.29		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 30 KD; no box for 0**
1.30		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 4; no box for 0**
pie charts		

1.31	$360 \div 30 = 12$ $11 \times 12 = 132$	**132°**
1.32	$360 \div 45 = 8$ $16 \times 8 = 128$	**128°**
1.33	$360 \div 60 = 6$ $25 \times 6 = 150$	**150°**
1.34	$360 \div 40 = 9$ $9 \times 9 = 81$	**81°**
1.35	$360 \div 72 = 5$ $20 \times 5 = 100$	**100°**
1.36	$360 \div 20 = 18$ $7 \times 18 = 126$	**126°**

Handling Data 4 · Averages and range (Lesson 4.5)

Q	Working	Answer
mean and range		
2.1	**(a)** $7 + 3 + 9 + 5 + 6 = 30$ $30 \div 5 = 6$ **(b)** $9 - 3 = 6$	**(a) 6** **(b) 6**
2.2	**(a)** $12 + 8 + 15 + 10 + 5 = 50$ $50 \div 5 = 10$ **(b)** $15 - 5 = 10$	**(a) 10** **(b) 10**
2.3	**(a)** $2 + 9 + 4 + 7 + 3 = 25$ $25 \div 5 = 5$ **(b)** $9 - 2 = 7$	**(a) 5** **(b) 7**
2.4	**(a)** $6 + 11 + 8 + 10 + 5 = 40$ $40 \div 5 = 8$ **(b)** $11 - 5 = 6$	**(a) 8** **(b) 6**

2.5	**(a)** $14 + 9 + 12 + 7 + 8 = 50$ $50 \div 5 = 10$ **(b)** $14 - 7 = 7$	**(a) 10** **(b) 7**
2.6	**(a)** $3 + 10 + 5 + 8 + 4 = 30$ $30 \div 5 = 6$ **(b)** $10 - 3 = 7$	**(a) 6** **(b) 7**
mode and median		
2.7	**(a)** 4 is the most common **(b)** $4, 4, 4, 5, 6, 8, 9$ $\text{middle} = 5$	**(a) 4** **(b) 5**
2.8	**(a)** 3 is the most common **(b)** $3, 3, 3, 5, 6, 7, 9$ $\text{middle} = 5$	**(a) 3** **(b) 5**
2.9	**(a)** 8 is the most common **(b)** $1, 2, 4, 5, 8, 8, 8$ $\text{middle} = 5$	**(a) 8** **(b) 5**
2.10	**(a)** 7 is the most common **(b)** $7, 7, 7, 9, 10, 11, 12$ $\text{middle} = 9$	**(a) 7** **(b) 9**
2.11	**(a)** 2 is the most common **(b)** $2, 2, 2, 3, 5, 6, 9$ $\text{middle} = 3$	**(a) 2** **(b) 3**
2.12	**(a)** 15 is the most common **(b)** $11, 12, 14, 15, 15, 15, 18$ $\text{middle} = 15$	**(a) 15** **(b) 15**